14

$\mathcal{H}_2$ Optimal Control

In this chapter we treat the optimal control of linear time-invariant systems with a quadratic performance criterion. The material in this chapter is standard, but the treatment is somewhat novel and lays the foundation for the subsequent chapters on $\mathcal{H}_\infty$-optimal control.

14.1 Introduction to Regulator Problem

Consider the following dynamical system:

$$\dot{x} = Ax + B_2 u, \quad x(t_0) = x_0 \tag{14.1}$$

where x_0 is given but arbitrary. Our objective is to find a control function $u(t)$ defined on $[t_0, T]$ which can be a function of the state $x(t)$ such that the state $x(t)$ is driven to a (small) neighborhood of origin at time T. This is the so-called *Regulator Problem*. One might suggest that this regulator problem can be trivially solved for any $T > t_0$ if the system is controllable. This is indeed the case if the controller can provide arbitrarily large amounts of energy since, by the definition of controllability, one can immediately construct a control function that will drive the state to zero in an arbitrarily short time. However, this is not practical since any physical system has the energy limitation, i.e., the actuator will eventually saturate. Furthermore, large control action can easily drive the system out of the region where the given linear model is valid. Hence certain limitations have to be imposed on the control in practical engineering implementation.

The constraints on control u may be measured in many different ways; for example,

$$\int_{t_0}^{T} \|u\|\, dt, \quad \int_{t_0}^{T} \|u\|^2\, dt, \quad \sup_{t\in[t_0,T]} \|u\|$$

i.e., in terms of $\mathcal{L}_1$-norm, $\mathcal{L}_2$-norm, and $\mathcal{L}_\infty$-norm, or more generally, weighted $\mathcal{L}_1$-norm, $\mathcal{L}_2$-norm, and $\mathcal{L}_\infty$-norm

$$\int_{t_0}^{T} \|W_u u\|\, dt, \quad \int_{t_0}^{T} \|W_u u\|^2\, dt, \quad \sup_{t\in[t_0,T]} \|W_u u\|$$

for some constant weighting matrix W_u.

Similarly, one might also want to impose some constraints on the transient response $x(t)$ in a similar fashion

$$\int_{t_0}^{T} \|W_x x\|\, dt, \quad \int_{t_0}^{T} \|W_x x\|^2\, dt, \quad \sup_{t\in[t_0,T]} \|W_x x\|$$

for some weighting matrix W_x. Hence the regulator problem can be posed as an optimal control problem with certain combined performance index on u and x, as given above. In this chapter, we shall be concerned exclusively with the $\mathcal{L}_2$ performance problem or quadratic performance problem. Moreover, we will focus on the infinite time regulator problem, i.e., $T \to \infty$, and, without loss of generality, we shall assume $t_0 = 0$. In this case, our problem is as follows: find a control $u(t)$ defined on $[0,\infty)$ such that the state $x(t)$ is driven to the origin at $t \to \infty$ and the following performance index is minimized:

$$\min_u \int_0^\infty \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}^* \begin{bmatrix} Q & S \\ S^* & R \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix} dt \tag{14.2}$$

for some $Q = Q^*$, S, and $R = R^* > 0$. This problem is traditionally called a *Linear Quadratic Regulator* problem or simply an LQR problem. Here we have assumed $R > 0$ to emphasis that the control energy has to be finite, i.e., $u(t) \in \mathcal{L}_2[0,\infty)$. So this is the space over which the integral is minimized. Moreover, it is also generally assumed that

$$\begin{bmatrix} Q & S \\ S^* & R \end{bmatrix} \geq 0. \tag{14.3}$$

Since R is positive definite, it has a square root, $R^{1/2}$, which is also positive-definite. By the substitution

$$u \leftarrow R^{1/2} u,$$

we may as well assume at the start that $R = I$. In fact, we can even assume $S = 0$ by using a pre-state feedback $u = -S^* x + v$ provided some care is exercised; however, this will not be assumed in the sequel. Since the matrix in (14.3) is positive semi-definite with $R = I$, it can be factored as

$$\begin{bmatrix} Q & S \\ S^* & I \end{bmatrix} = \begin{bmatrix} C_1^* \\ D_{12}^* \end{bmatrix} \begin{bmatrix} C_1 & D_{12} \end{bmatrix}.$$

And (14.2) can be rewritten as

$$\min_{u \in \mathcal{L}_2[0,\infty)} \|C_1 x + D_{12} u\|_2^2 .$$

In fact, the LQR problem is posed traditionally as the minimization problem

$$\min_{u \in \mathcal{L}_2[0,\infty)} \|C_1 x + D_{12} u\|_2^2 \tag{14.4}$$

$$\dot{x} = Ax + Bu, \quad x(0) = x_0 \tag{14.5}$$

without explicitly mentioning the condition that the control should drive the state to the origin. Instead some assumptions are imposed on Q, S, and R (or equivalently C_1 and D_{12}) to ensure that the optimal control law u has this property. To see what assumption one needs to make in order to ensure that the minimization problem formulated in (14.4) and (14.5) has a sensible solution, let us consider a simple example with $A = 1, B = 1, Q = 0, S = 0$, and $R = 1$:

$$\min_{u \in \mathcal{L}_2[0,\infty)} \int_0^\infty u^2 dt, \quad \dot{x} = x + u, \quad x(0) = x_0.$$

It is clear that $u = 0$ is the optimal solution. However, the system with $u = 0$ is unstable and $x(t)$ diverges exponentially to infinity, $x(t) = e^t x_0$. The problem with this example is that the performance index does not "see" the unstable state x. This is true in general, and the proof of this fact is left as an exercise to the reader. Hence in order to ensure that the minimization problem in (14.4) and (14.5) is sensible, we must assume that all unstable states can be "seen" from the performance index, i.e., (C_1, A) must be detectable. This will be called a *standard LQR problem*.

On the other hand, if the closed-loop stability is imposed on the above minimization, then it can be shown that $\min_{u \in \mathcal{L}_2[0,\infty)} \int_0^\infty u^2 dt = 2x_0^2$ and $u(t) = -2x(t)$ is the optimal control. This can also be generalized to a more general case where (C_1, A) is not necessarily detectable. This problem will be referred to as an *Extended LQR problem*.

14.2 Standard LQR Problem

In this section, we shall consider the LQR problem as traditionally formulated.

Standard LQR Problem

Let a dynamical system be described by

$$\dot{x} = Ax + B_2 u, \quad x(0) = x_0 \text{ given but arbitrary} \tag{14.6}$$
$$z = C_1 x + D_{12} u \tag{14.7}$$

and suppose that the system parameter matrices satisfy the following assumptions:

(A1) (A, B_2) is stabilizable;

(A2) D_{12} has full column rank with $\begin{bmatrix} D_{12} & D_\perp \end{bmatrix}$ unitary;

(A3) (C_1, A) is detectable;

(A4) $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all ω.

Find an optimal control law $u \in \mathcal{L}_2[0,\infty)$ such that the performance criterion $\|z\|_2^2$ is minimized.

Remark 14.1 Assumption (A1) is clearly necessary for the existence of a stabilizing control function u. The assumption (A2) is made for simplicity of notation and is actually a restatement that $R = D_{12}^* D_{12} = I$. Note also that $D_\perp$ drops out when D_{12} is square. It is interesting to point out that (A3) is not needed in the Extended LQR problem. The assumption (A3) enforces that the unconditional optimization problem will result in a stabilizing control law. In fact, the assumption (A3) together with (A1) guarantees that the input/output stability implies the internal stability, i.e., $u \in \mathcal{L}_2$ and $z \in \mathcal{L}_2$ imply $x \in \mathcal{L}_2$, which will be shown in Lemma 14.1. Finally note that (A4) is equivalent to the condition that $(D_\perp^* C_1, A - B_2 D_{12}^* C_1)$ has no unobservable modes on the imaginary axis and is weaker than the popular assumption of detectability of $(D_\perp^* C_1, A - B_2 D_{12}^* C_1)$. (A4), together with the stabilizability of (A, B_2), guarantees by Corollary 13.10 that the following Hamiltonian matrix belongs to $dom(Ric)$ and that $X = Ric(H) \geq 0$:

$$\begin{aligned} H &= \begin{bmatrix} A & 0 \\ -C_1^* C_1 & -A^* \end{bmatrix} - \begin{bmatrix} B_2 \\ -C_1^* D_{12} \end{bmatrix} \begin{bmatrix} D_{12}^* C_1 & B_2^* \end{bmatrix} \\ &= \begin{bmatrix} A - B_2 D_{12}^* C_1 & -B_2 B_2^* \\ -C_1^* D_\perp D_\perp^* C_1 & -(A - B_2 D_{12}^* C_1)^* \end{bmatrix}. \end{aligned} \tag{14.8}$$

Note also that if $D_{12}^* C_1 = 0$, then (A4) is implied by the detectability of (C_1, A), while the detectability of (C_1, A) is implied by the detectability of $(D_\perp^* C_1, A - B_2 D_{12}^* C_1)$. The above implication is not true if $D_{12}^* C_1 \neq 0$, for example,

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D_{12} = 1.$$

Then (C_1, A) is detectable and $A - B_2 D_{12}^* C_1 = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$ has no eigenvalue on the imaginary axis but is not stable. ♡

Note also that the Riccati equation corresponding to (14.8) is

$$(A - B_2 D_{12}^* C_1)^* X + X(A - B_2 D_{12}^* C_1) - X B_2 B_2^* X + C_1^* D_\perp D_\perp^* C_1 = 0. \tag{14.9}$$

Now let X be the corresponding stabilizing solution and define

$$F := -(B_2^* X + D_{12}^* C_1). \tag{14.10}$$

Then $A + B_2 F$ is stable. Denote

$$A_F := A + B_2 F, \quad C_F := C_1 + D_{12} F$$

and re-arrange equation (14.9) to get

$$A_F^* X + X A_F + C_F^* C_F = 0. \tag{14.11}$$

Thus X is the observability Gramian of (C_F, A_F).

Consider applying the control law $u = Fx$ to the system (14.6) and (14.7). The controlled system is

$$\begin{aligned} \dot{x} &= A_F x, \quad x(0) = x_0 \\ z &= C_F x \end{aligned}$$

or equivalently

$$\begin{aligned} \dot{x} &= A_F x + x_0 \delta(t), \quad x(0_-) = 0 \\ z &= C_F x. \end{aligned}$$

The associated transfer matrix is

$$G_c(s) = \left[\begin{array}{c|c} A_F & I \\ \hline C_F & 0 \end{array}\right]$$

and

$$\|G_c x_0\|_2^2 = x_0^* X x_0.$$

The proof of the following theorem requires a preliminary result about internal stability given input-output stability.

Lemma 14.1 *If $u, z \in \mathcal{L}_p[0, \infty)$ for $p \geq 1$ and (C_1, A) is detectable in the system described by equations (14.6) and (14.7), then $x \in \mathcal{L}_p[0, \infty)$. Furthermore, if $p < \infty$, then $x(t) \to 0$ as $t \to \infty$.*

Proof. Since (C_1, A) is detectable, there exists L such that $A + LC_1$ is stable. Let $\hat{x}$ be the state estimate of x given by

$$\dot{\hat{x}} = (A + LC_1)\hat{x} + (LD_{12} + B_2)u - Lz.$$

Then $\hat{x} \in \mathcal{L}_p[0, \infty)$ since z and u are in $\mathcal{L}_p[0, \infty)$. Now let $e = x - \hat{x}$; then

$$\dot{e} = (A + LC_1)e$$

and $e \in \mathcal{L}_p[0, \infty)$. Therefore, $x = e + \hat{x} \in \mathcal{L}_p[0, \infty)$. It is easy to see that $e(t) \to 0$ as $t \to \infty$ for any initial condition $e(0)$. Finally, $x(t) \to 0$ follows from the fact that if $p < \infty$ then $\hat{x} \to 0$. □

Theorem 14.2 *There exists a unique optimal control for the LQR problem, namely $u = Fx$. Moreover,*

$$\min_{u \in \mathcal{L}_2[0,\infty)} \|z\|_2 = \|G_c x_0\|_2 .$$

Note that the optimal control strategy is constant gain state feedback, and this gain is independent of the initial condition x_0.

Proof. With the change of variable $v = u - Fx$, the system can be written as

$$\begin{bmatrix} \dot{x} \\ z \end{bmatrix} = \begin{bmatrix} A_F & B_2 \\ C_F & D_{12} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix} , \qquad x(0) = x_0. \tag{14.12}$$

Now if $v \in \mathcal{L}_2[0, \infty)$, then $x, z \in \mathcal{L}_2[0, \infty)$ and $x(\infty) = 0$ since A_F is stable. Hence $u = Fx + v \in \mathcal{L}_2[0, \infty)$. Conversely, if $u, z \in \mathcal{L}_2[0, \infty)$, then from Lemma 14.1 $x \in \mathcal{L}_2[0, \infty)$. So $v \in \mathcal{L}_2[0, \infty)$. Thus the mapping $v = u - Fx$ between $v \in \mathcal{L}_2[0, \infty)$ and those $u \in \mathcal{L}_2[0, \infty)$ that make $z \in \mathcal{L}_2[0, \infty)$ is one-to-one and onto. Therefore,

$$\min_{u \in \mathcal{L}_2[0,\infty)} \|z\|_2 = \min_{v \in \mathcal{L}_2[0,\infty)} \|z\|_2 .$$

By differentiating $x(t)^* X x(t)$ with respect to t along a solution of the differential equation (14.12) and by using (14.9) and the fact that $C_F^* D_{12} = -XB_2$, we see that

$$\begin{aligned} \frac{d}{dt} x^* X x &= \dot{x}^* X x + x^* X \dot{x} \\ &= x^*(A_F^* X + X A_F)x + 2x^* X B_2 v \\ &= -x^* C_F^* C_F x + 2x^* X B_2 v \\ &= -(C_F x + D_{12} v)^*(C_F x + D_{12} v) + 2x^* C_F^* D_{12} v + v^* v + 2x^* X B_2 v \\ &= -\|z\|^2 + \|v\|^2 . \end{aligned} \tag{14.13}$$

Now integrate (14.13) from 0 to ∞ to get

$$\|z\|_2^2 = x_0^* X x_0 + \|v\|_2^2 .$$

Clearly, the unique optimal control is $v = 0$, i.e., $u = Fx$. $\square$

This method of proof, involving change of variables and the completion of the square, is a standard technique and variants of it will be used throughout this book. An alternative proof can be given in frequency domain. To do that, let us first note the following fact:

Lemma 14.3 *Let a transfer matrix be defined as*

$$U := \left[\begin{array}{c|c} A_F & B_2 \\ \hline C_F & D_{12} \end{array}\right] \in \mathcal{RH}_\infty .$$

Then U is inner and $U^\sim G_c \in \mathcal{RH}_2^\perp$.

Proof. The proof uses standard manipulations of state space realizations. From U we get

$$U^\sim(s) = \left[\begin{array}{c|c} -A_F^* & -C_F^* \\ \hline B_2^* & D_{12}^* \end{array}\right].$$

Then it is easy to compute

$$U^\sim U = \left[\begin{array}{cc|c} -A_F^* & -C_F^* C_F & -C_F^* D_{12} \\ 0 & A_F & B_2 \\ \hline B_2^* & D_{12}^* C_F & I \end{array}\right], \quad U^\sim G_c = \left[\begin{array}{cc|c} -A_F^* & -C_F^* C_F & 0 \\ 0 & A_F & I \\ \hline B_2^* & D_{12}^* C_F & 0 \end{array}\right].$$

Now do the similarity transformation

$$\begin{bmatrix} I & -X \\ 0 & I \end{bmatrix}$$

on the states of the transfer matrices and use (14.11) to get

$$U^\sim U = \left[\begin{array}{cc|c} -A_F^* & 0 & 0 \\ 0 & A_F & B_2 \\ \hline B_2^* & 0 & I \end{array}\right] = I$$

$$U^\sim G_c = \left[\begin{array}{cc|c} -A_F^* & 0 & -X \\ 0 & A_F & I \\ \hline B_2^* & 0 & 0 \end{array}\right] = \left[\begin{array}{c|c} -A_F^* & -X \\ \hline B_2^* & 0 \end{array}\right] \in \mathcal{RH}_2^\perp .$$

□

An alternative proof of Theorem 14.2 We have in the frequency domain

$$z = G_c x_0 + Uv.$$

Let $v \in \mathcal{H}_2$. By Lemma 14.3, $G_c x_0$ and Uv are orthogonal. Hence

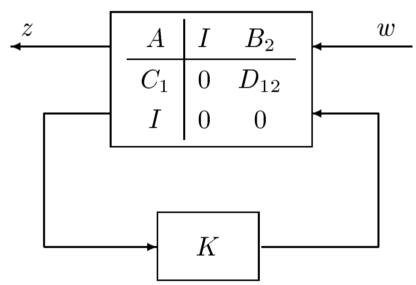
$$\|z\|_2^2 = \|G_c x_0\|_2^2 + \|Uv\|_2^2 .$$

Since U is inner, we get

$$\|z\|_2^2 = \|G_c x_0\|_2^2 + \|v\|_2^2 .$$

This equation immediately gives the desired conclusion. □

Remark 14.2 It is clear that the LQR problem considered above is essentially equivalent to minimizing the 2-norm of z with the input $w = x_0 \delta(t)$ in the following diagram:



But this problem is a special $\mathcal{H}_2$ norm minimization problem considered in a later section. ♡

14.3 Extended LQR Problem

This section considers the extended LQR problem where no detectability assumption is made for (C_1, A).

Extended LQR Problem

Let a dynamical system be given by

$$\begin{aligned} \dot{x} &= Ax + B_2 u, \quad x(0) = x_0 \text{ given but arbitrary} \\ z &= C_1 x + D_{12} u \end{aligned}$$

with the following assumptions:

(A1) (A, B_2) is stabilizable;

(A2) D_{12} has full column rank with $\begin{bmatrix} D_{12} & D_\perp \end{bmatrix}$ unitary;

(A3) $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all ω.

Find an optimal control law $u \in \mathcal{L}_2[0,\infty)$ such that the system is internally stable, i.e., $x \in \mathcal{L}_2[0,\infty)$ and the performance criterion $\|z\|_2^2$ is minimized.

Assume the same notation as above, and we have

Theorem 14.4 *There exists a unique optimal control for the extended LQR problem, namely $u = Fx$. Moreover,*

$$\min_{u \in \mathcal{L}_2[0,\infty)} \|z\|_2 = \|G_c x_0\|_2 .$$

Proof. The proof of this theorem is very similar to the proof of the standard LQR problem except that, in this case, the input/output stability may not necessarily imply the internal stability. Instead, the internal stability is guaranteed by the way of choosing control law.

Suppose that $u \in \mathcal{L}_2[0,\infty)$ is such a control law that the system is stable, i.e., $x \in \mathcal{L}_2[0,\infty)$. Then $v = u - Fx \in \mathcal{L}_2[0,\infty)$. On the other hand, let $v \in \mathcal{L}_2[0,\infty)$ and consider

$$\begin{bmatrix} \dot{x} \\ z \end{bmatrix} = \begin{bmatrix} A_F & B_2 \\ C_F & D_{12} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}, \qquad x(0) = x_0.$$

Then $x, z \in \mathcal{L}_2[0,\infty)$ and $x(\infty) = 0$ since A_F is stable. Hence $u = Fx + v \in \mathcal{L}_2[0,\infty)$. Again the mapping $v = u - Fx$ between $v \in \mathcal{L}_2[0,\infty)$ and those $u \in \mathcal{L}_2[0,\infty)$ that make $z \in \mathcal{L}_2[0,\infty)$ and $x \in \mathcal{L}_2[0,\infty)$ is one to one and onto. Therefore,

$$\min_{u \in \mathcal{L}_2[0,\infty)} \|z\|_2 = \min_{v \in \mathcal{L}_2[0,\infty)} \|z\|_2 .$$

Using the same technique as in the proof of the standard LQR problem, we have

$$\|z\|_2^2 = x_0^* X x_0 + \|v\|_2^2 .$$

And the unique optimal control is $v = 0$, i.e., $u = Fx$. □

14.4 Guaranteed Stability Margins of LQR

Now we will consider the system described by equation (14.6) with the LQR control law $u = Fx$. The closed-loop block diagram is as shown in Figure 14.1.

The following result is the key to stability margins of an LQR control law.

Lemma 14.5 *Let $F = -(B_2^* X + D_{12}^* C_1)$ and define $G_{12} = D_{12} + C_1(sI - A)^{-1}B_2$. Then*

$$\left(I - B_2^*(-sI - A^*)^{-1}F^*\right)\left(I - F(sI - A)^{-1}B_2\right) = G_{12}^\sim(s) G_{12}(s).$$

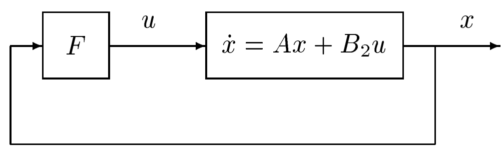


Figure 14.1: LQR closed-loop system

Proof. Note that the Riccati equation (14.9) can be written as

$$XA + A^*X - F^*F + C_1^*C_1 = 0.$$

Add and subtract sX to the equation to get

$$-X(sI - A) - (-sI - A^*)X - F^*F + C_1^*C_1 = 0.$$

Now multiply the above equation from the left by $B_2^*(-sI - A^*)^{-1}$ and from the right by $(sI - A)^{-1}B_2$ to get

$$-B_2^*(-sI - A^*)^{-1}XB_2 - B_2^*X(sI - A)^{-1}B_2 - B_2^*(-sI - A^*)^{-1}F^*F(sI - A)^{-1}B_2$$
$$+B_2^*(-sI - A^*)^{-1}C_1^*C_1(sI - A)^{-1}B_2 = 0.$$

Using $-B_2^*X = F + D_{12}^*C_1$ in the above equation, we have

$$B_2^*(-sI - A^*)^{-1}F^* + F(sI - A)^{-1}B_2 - B_2^*(-sI - A^*)^{-1}F^*F(sI - A)^{-1}B_2$$
$$+B_2^*(-sI - A^*)^{-1}C_1^*D_{12} + D_{12}^*C_1(sI - A)^{-1}B_2$$
$$+B_2^*(-sI - A^*)^{-1}C_1^*C_1(sI - A)^{-1}B_2 = 0.$$

Then the result follows from completing the square and from the fact that $D_{12}^*D_{12} = I$. $\square$

Corollary 14.6 *Suppose* $D_{12}^*C_1 = 0$. *Then*

$$\left(I - B_2^*(-sI - A^*)^{-1}F^*\right)\left(I - F(sI - A)^{-1}B_2\right) = I + B_2^*(-sI - A^*)^{-1}C_1^*C_1(sI - A)^{-1}B_2.$$

In particular,

$$\left(I - B_2^*(-j\omega I - A^*)^{-1}F^*\right)\left(I - F(j\omega I - A)^{-1}B_2\right) \geq I \tag{14.14}$$

and

$$\left(I + B_2^*(-j\omega I - A^* - F^*B_2^*)^{-1}F^*\right)\left(I + F(j\omega I - A - B_2F)^{-1}B_2\right) \leq I. \tag{14.15}$$

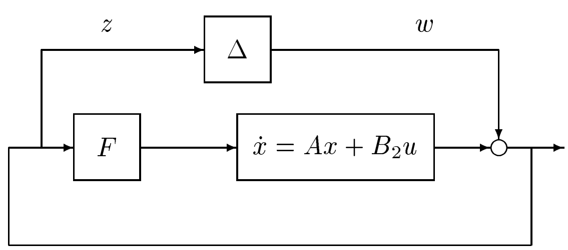
Note that the inequality (14.15) follows from taking the inverse of inequality (14.14).

Define $G(s) = -F(sI - A)^{-1}B_2$ and assume for the moment that the system is single input. Then the inequality (14.14) shows that the open-loop Nyquist diagram of the system $G(s)$ in Figure 14.1 never enters the unit disk centered at $(-1, 0)$ of the complex plane. Hence the system has at least the following stability margins:

$$k_{\min} \le \frac{1}{2}, \quad k_{\max} = \infty, \quad \phi_{\min} \le -60^o, \quad \phi_{\max} \ge 60^o$$

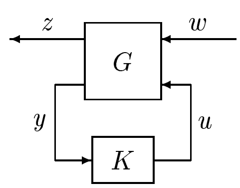
i.e., the system has at least a $6dB$ ($= 20\log 2$) gain margin and a 60^o phase margin in both directions. A similar interpretation may be generalized to multiple input systems.

Next, it is noted that the inequality (14.15) can also be given some robustness interpretation. In fact, it implies that the closed-loop system in Figure 14.1 is stable even if the open-loop system $G(s)$ is perturbed additively by a $\Delta \in \mathcal{RH}_\infty$ as long as $\|\Delta\|_\infty < 1$. This can be seen from the following block diagram and small gain theorem where the transfer matrix from w to z is exactly $I + F(j\omega I - A - B_2F)^{-1}B_2$.



14.5 Standard $\mathcal{H}_2$ Problem

The system considered in this section is described by the following standard block diagram:



The realization of the transfer matrix G is taken to be of the form

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{array}\right].$$

Notice the special off-diagonal structure of D: D_{22} is assumed to be zero so that G_{22} is strictly proper[1]; also, D_{11} is assumed to be zero in order to guarantee that the $\mathcal{H}_2$ problem properly posed.[2] The case for $D_{11} \neq 0$ will be discussed in Section 14.7.

The following additional assumptions are made for the output feedback $\mathcal{H}_2$ problem in this chapter:

(i) (A, B_2) is stabilizable and (C_2, A) is detectable;

(ii) D_{12} has full column rank with $\begin{bmatrix} D_{12} & D_\perp \end{bmatrix}$ unitary, and D_{21} has full row rank with $\begin{bmatrix} D_{21} \\ \tilde{D}_\perp \end{bmatrix}$ unitary;

(iii) $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all ω;

(iv) $\begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix}$ has full row rank for all ω.

The first assumption is for the stabilizability of G by output feedback, and the third and the fourth assumptions together with the first guarantee that the two Hamiltonian matrices associated with the $\mathcal{H}_2$ problem below belong to *dom*(*Ric*). The rank assumptions (ii) are necessary to guarantee that the $\mathcal{H}_2$ optimal controller is a finite dimensional linear time invariant one, while the unitary assumptions are made for the simplicity of the final solution; they are not restrictions (see e.g., Chapter 17).

> $\mathcal{H}_2$ **Problem** *The $\mathcal{H}_2$ control problem is to find a proper, real-rational controller K which stabilizes G internally and minimizes the $\mathcal{H}_2$-norm of the transfer matrix T_{zw} from w to z.*

In the following discussions we shall assume that we have state models of G and K. Recall that a controller is said to be admissible if it is internally stabilizing and proper.

We now state the solution of the problem and then take up its derivation in the next several sections. By Corollary 13.10 the two Hamiltonian matrices

$$H_2 := \begin{bmatrix} A & 0 \\ -C_1^* C_1 & -A^* \end{bmatrix} - \begin{bmatrix} B_2 \\ -C_1^* D_{12} \end{bmatrix} \begin{bmatrix} D_{12}^* C_1 & B_2^* \end{bmatrix}$$

$$J_2 := \begin{bmatrix} A^* & 0 \\ -B_1 B_1^* & -A \end{bmatrix} - \begin{bmatrix} C_2^* \\ -B_1 D_{21}^* \end{bmatrix} \begin{bmatrix} D_{21} B_1^* & C_2 \end{bmatrix}$$

[1] As we have discussed in Section 12.3.4 of Chapter 12 there is no loss of generality in making this assumption since the controller for D_{22} nonzero case can be recovered from the zero case.

[2] Recall that a rational proper stable transfer function is an $\mathcal{RH}_2$ function iff it is strictly proper.

belong to $dom(Ric)$, and, moreover, $X_2 := Ric(H_2) \geq 0$ and $Y_2 := Ric(J_2) \geq 0$. Define

$$F_2 := -(B_2^* X_2 + D_{12}^* C_1), \quad L_2 := -(Y_2 C_2^* + B_1 D_{21}^*)$$

and

$$A_{F_2} := A + B_2 F_2, \quad C_{1F_2} := C_1 + D_{12} F_2$$

$$A_{L_2} := A + L_2 C_2, \quad B_{1L_2} := B_1 + L_2 D_{21}$$

$$\hat{A}_2 := A + B_2 F_2 + L_2 C_2$$

$$G_c(s) := \left[\begin{array}{c|c} A_{F_2} & I \\ \hline C_{1F_2} & 0 \end{array}\right], \qquad G_f(s) := \left[\begin{array}{c|c} A_{L_2} & B_{1L_2} \\ \hline I & 0 \end{array}\right].$$

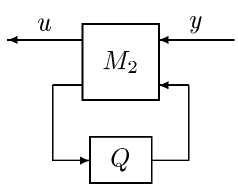
Theorem 14.7 *There exists a unique optimal controller*

$$K_{opt}(s) := \left[\begin{array}{c|c} \hat{A}_2 & -L_2 \\ \hline F_2 & 0 \end{array}\right].$$

Moreover, $\min \|T_{zw}\|_2^2 = \|G_c B_1\|_2^2 + \|F_2 G_f\|_2^2 = \|G_c L_2\|_2^2 + \|C_1 G_f\|_2^2$.

The controller K_{opt} has the well-known separation structure, which will be discussed in more detail in Section 14.9. For comparison with the $\mathcal{H}_\infty$ results, it is useful to describe all suboptimal controllers.

Theorem 14.8 *The family of all admissible controllers such that $\|T_{zw}\|_2 < \gamma$ equals the set of all transfer matrices from y to u in*



$$M_2(s) = \left[\begin{array}{c|cc} \hat{A}_2 & -L_2 & B_2 \\ \hline F_2 & 0 & I \\ -C_2 & I & 0 \end{array}\right]$$

where $Q \in \mathcal{RH}_2$, $\|Q\|_2^2 < \gamma^2 - (\|G_c B_1\|_2^2 + \|F_2 G_f\|_2^2)$.

Thus, the suboptimal controllers are parameterized by a fixed (independent of γ) linear-fractional transformation with a free parameter Q. With $Q = 0$, we recover K_{opt}. It is worth noting that the parameterization in Theorem 14.8 makes T_{zw} affine in Q and yields the Youla parameterization of all stabilizing controllers when the conditions on Q are replaced by $Q \in \mathcal{RH}_\infty$.

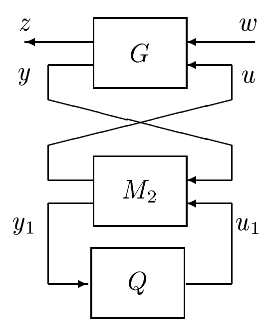
14.6 Optimal Controlled System

In this section, we look at the controller,

$$K(s) = \mathcal{F}_\ell(M_2, Q), \quad Q \in \mathcal{RH}_\infty$$

connected to G. (Keep in mind that all admissible controllers are parameterized by the above formula). We will give a brief analysis of the closed-loop system. It will be seen that a direct consequence from this analysis is the results of Theorem 14.7 and 14.8. The proof given here is not our emphasis. The reason is that this approach does not generalize nicely to other control problems and is often very involved. An alternative proof will be given in the later part of this chapter by using the FI and OE results discussed in section 14.8 and the separation argument. The idea of separation is the main theme for synthesis. We shall now analyze the system structure under the control of a such controller. In particular, we will compute explicitly $\|T_{zw}\|_2^2$.

Consider the following system diagram with controller $K(s) = \mathcal{F}_\ell(M_2, Q)$:



Then $T_{zw} = \mathcal{F}_\ell(N, Q)$ with

$$N = \left[\begin{array}{cc|cc} A_{F_2} & -B_2F_2 & B_1 & B_2 \\ 0 & A_{L_2} & B_{1L_2} & 0 \\ \hline C_{1F_2} & -D_{12}F_2 & 0 & D_{12} \\ 0 & C_2 & D_{21} & 0 \end{array}\right].$$

Define

$$U = \left[\begin{array}{c|c} A_{F_2} & B_2 \\ \hline C_{1F_2} & D_{12} \end{array}\right], \quad V = \left[\begin{array}{c|c} A_{L_2} & B_{1L_2} \\ \hline C_2 & D_{21} \end{array}\right].$$

We have

$$T_{zw} = G_cB_1 - UF_2G_f + UQV.$$

It follows from Lemma 14.3 that G_cB_1 and U are orthogonal. Thus

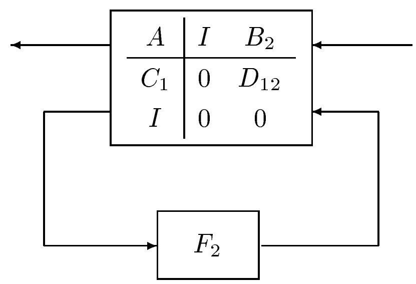
$$\|T_{zw}\|_2^2 = \|G_cB_1\|_2^2 + \|UF_2G_f - UQV\|_2^2 = \|G_cB_1\|_2^2 + \|F_2G_f - QV\|_2^2.$$

It can also be shown easily by duality that G_f and V are orthogonal, i.e., $G_f V^{\sim} \in \mathcal{RH}_2^{\perp}$, and V is a co-inner, so we have

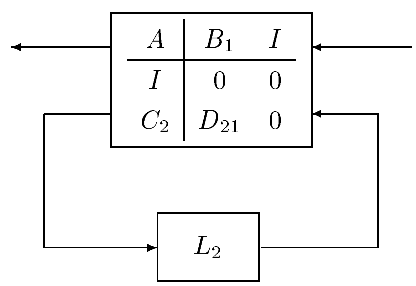
$$\|T_{zw}\|_2^2 = \|G_c B_1\|_2^2 + \|F_2 G_f - QV\|_2^2 = \|G_c B_1\|_2^2 + \|F_2 G_f\|_2^2 + \|Q\|_2^2 .$$

This shows clearly that $Q = 0$ gives the unique optimal control, so $K = \mathcal{F}_\ell(M_2, 0)$ is the unique optimal controller. Note also that $\|T_{zw}\|_2$ is finite if and only if $Q \in \mathcal{RH}_2$. Hence Theorem 14.7 and 14.8 follow easily.

It is interesting to examine the structure of G_c and G_f. First of all the transfer matrix G_c can be represented as a fixed system with the feedback matrix F_2 wrapped around it:



F_2 is, in fact, an optimal LQR controller and minimizes the $\mathcal{H}_2$ norm of G_c. Similarly, G_f can be represented as



and L_2 minimizes the $\mathcal{H}_2$ norm of G_f and solves a special filtering problem.

14.7 $\mathcal{H}_2$ Control with Direct Disturbance Feedforward*

Let us consider the generalized system structure again with D_{11} not necessarily zero:

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & 0 \end{array}\right]$$

We shall consider the following question: what will happen and under what condition will the $\mathcal{H}_2$ optimal control problem make sense if $D_{11} \neq 0$?

Recall that $\mathcal{F}_\ell(M_2, Q)$ with $Q \in \mathcal{RH}_\infty$ parameterizes all stabilizing controllers for G regardless of $D_{11} = 0$ or not. Now again consider the closed loop transfer matrix with the controller $K = \mathcal{F}_\ell(M_2, Q)$; then

$$T_{zw} = G_c B_1 - U F_2 G_f + UQV + D_{11}$$

and

$$T_{zw}(\infty) = D_{12} Q(\infty) D_{21} + D_{11}.$$

Hence the $\mathcal{H}_2$ optimal control problem will make sense, i.e., having finite $\mathcal{H}_2$ norm, if and only if there is a constant $Q(\infty)$ such that

$$D_{12} Q(\infty) D_{21} + D_{11} = 0.$$

This requires that

$$Q(\infty) = -D_{12}^* D_{11} D_{21}^*$$

and that

$$-D_{12} D_{12}^* D_{11} D_{21}^* D_{21} + D_{11} = 0. \tag{14.16}$$

Note that the equation (14.16) is a very restrictive condition. For example, suppose

$$D_{12} = \begin{bmatrix} 0 \\ I \end{bmatrix}, \qquad D_{21} = \begin{bmatrix} 0 & I \end{bmatrix}$$

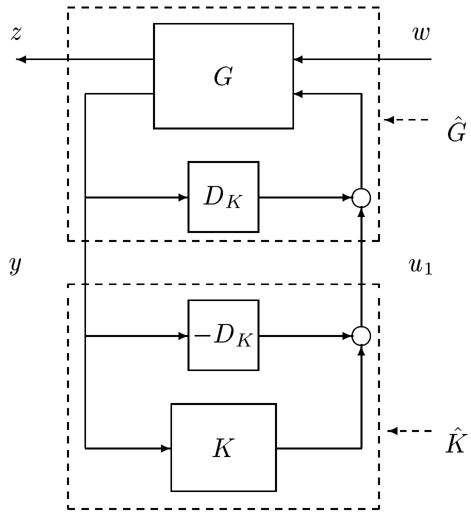
and D_{11} is partitioned accordingly

$$D_{11} = \begin{bmatrix} D_{1111} & D_{1112} \\ D_{1121} & D_{1122} \end{bmatrix}.$$

Then equation (14.16) implies that

$$\begin{bmatrix} D_{1111} & D_{1112} \\ D_{1121} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

and that $Q(\infty) = -D_{1122}$. So only D_{1122} can be nonzero for a sensible $\mathcal{H}_2$ problem. Hence from now on in this section we shall assume that (14.16) holds and denotes $D_K := -D_{12}^* D_{11} D_{21}^*$. To find the optimal control law for the system G with $D_{11} \neq 0$, let us consider the following system configuration:



Then

$$\hat{G} = \left[\begin{array}{c|cc} A + B_2 D_K C_2 & B_1 + B_2 D_K D_{21} & B_2 \\ \hline C_1 + D_{12} D_K C_2 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{array}\right]$$

and

$$K = D_K + \hat{K}.$$

It is easy to check that the system $\hat{G}$ satisfies all assumptions in Section 14.5; hence the controller formula in Section 14.5 can be used. A little bit of algebra will show that

$$\hat{K} = \left[\begin{array}{c|c} \hat{A}_2 - B_2 D_K C_2 & -(L_2 - B_2 D_K) \\ \hline F_2 - D_K C_2 & 0 \end{array}\right]$$

is the $\mathcal{H}_2$ optimal controller for $\hat{G}$. Hence the controller K for the original system G will be given by

$$K = \left[\begin{array}{c|c} \hat{A}_2 - B_2 D_K C_2 & -(L_2 - B_2 D_K) \\ \hline F_2 - D_K C_2 & D_K \end{array}\right] = \mathcal{F}_\ell(M_2, D_K).$$

14.8 Special Problems

In this section we look at various $\mathcal{H}_2$-optimization problems from which the output feedback solutions of the previous sections will be constructed via a separation argument. All the special problems in this section are to find K stabilizing G and minimizing the $\mathcal{H}_2$-norm from w to z in the standard setup, but with different structures for G. As in Chapter 12, we shall call these special problems, respectively, state feedback (SF), output injection (OI), full information (FI), full control (FC), disturbance feedforward (DF), and output estimation (OE). OI, FC, and OE are natural duals of SF, FI, and DF, respectively. The output feedback solutions will be constructed out of the FI and OE results.

The special problems SF, OI, FI, and FC are not, strictly speaking, special cases of the output feedback problem since they do not satisfy all of the assumptions for output feedback (while DF and OE do). Each special problem inherits some of the assumptions (i)-(iv) from the output feedback as appropriate. The assumptions will be discussed in the subsections for each problem.

In each case, the results are summarized as a list of three items; (in all cases, K must be admissible)

1. the minimum of $\|T_{zw}\|_2$;
2. the unique controller minimizing $\|T_{zw}\|_2$;
3. the family of all controllers such that $\|T_{zw}\|_2 < \gamma$, where γ is greater than the minimum norm.

Warning: we will be more specific below about what we mean about the uniqueness and all controllers in the second and third item. In particular, the controllers characterized here for SF, OI, FI and FC problems are neither unique nor all-inclusive. Once again we regard all controllers that give the same control signal u, i.e., having the same transfer function from w to u, as an equivalence class. In other words, if K_1 and K_2 generate the same control signal u, we will regard them as the same, denoted as $K_1 \cong K_2$. Hence the "unique controller" here means one of the controllers from the unique equivalence class. The same comments apply to the "all" situation. This will be much clearer in section 14.8.1 when we consider the state feedback problem. In that case we actually give a parameterization of all the elements in the equivalence class of the "unique" optimal controllers. Thus the unique controller is really not unique. We chose not to give the parameterization of all the elements in an equivalence class in this book since it is very messy, as can be seen in section 14.8.1, and not very useful. However, it will be seen that this equivalence class problem will not occur in the general output feedback case including DF and OE problems.

14.8.1 State Feedback

Consider an open-loop system transfer matrix

$$G_{SF}(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ I & 0 & 0 \end{array}\right]$$

with the following assumptions:

(i) (A, B_2) is stabilizable;

(ii) D_{12} has full column rank with $\begin{bmatrix} D_{12} & D_\perp \end{bmatrix}$ unitary;

(iii) $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all ω.

This is very much like the LQR problem except that we require from the start that u be generated by state feedback and that the detectability of (C_1, A) is not imposed since the controllers are restricted to providing internal stability. The controller is allowed to be dynamic, but it turns out that dynamics are not necessary.

State Feedback:

1. $\min \|T_{zw}\|_2 = \|G_c B_1\|_2 = (\mathrm{trace}(B_1^* X_2 B_1))^{1/2}$

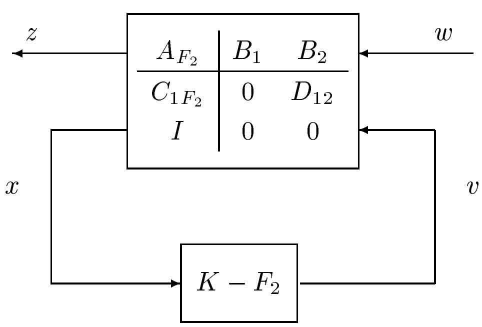
2. $K(s) \cong F_2$

Remark 14.3 The class of all suboptimal controllers for state feedback are messy and are not very useful in this book, so they are omitted, as are the OI problems. ♡

Proof. Let K be a stabilizing controller, $u = K(s)x$. Change control variables by defining $v := u - F_2 x$ and then write the system equations as

$$\begin{bmatrix} \dot{x} \\ z \\ v \end{bmatrix} = \begin{bmatrix} A_{F_2} & B_1 & B_2 \\ C_{1F_2} & 0 & D_{12} \\ (K - F_2) & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ w \\ v \end{bmatrix}.$$

The block diagram is



Let T_{vw} denote the transfer matrix from w to v. Notice that $T_{vw} \in \mathcal{RH}_2$ because K stabilizes G. Then

$$T_{zw} = G_c B_1 + U T_{vw}$$

where $U = \left[\begin{array}{c|c} A_{F_2} & B_2 \\ \hline C_{1F_2} & D_{12} \end{array}\right]$, and by Lemma 14.3 U is inner and $U^{\sim}G_c$ is in $\mathcal{RH}_2^{\perp}$. We get

$$\|T_{zw}\|_2^2 = \|G_c B_1\|_2^2 + \|T_{vw}\|_2^2 .$$

Thus $\min \|T_{zw}\|_2^2 = \|G_c B_1\|_2^2$ and the minimum is achieved iff $T_{vw} = 0$. Furthermore, $K = F_2$ is a controller achieving this minimum, and any other controllers achieving minimum are in the equivalence class of F_2. □

Note that the above proof actually yields a much stronger result than what is needed. The proof that the optimal T_{vw} is $T_{vw} = 0$ does not depend on the restriction that the controller measures just the state. We only require that the controller produce v as a causal stable function T_{vw} of w. This means that the optimal state feedback is also optimal for the full information problem as well.

We now give some further explanation about the uniqueness of the optimal controller that we commented on before. The important observation for this issue is that the controllers making $T_{vw} = 0$ are not unique. The controller given above, F_2, is only one of them. We will now try to find all of those controllers that stabilize the system and give $T_{vw} = 0$, i.e., all $K(s) \cong F_2$.

Proposition 14.9 *Let V_c be a matrix whose columns form a basis for $KerB_1^*$ ($V_c^* B_1 = 0$). Then all $\mathcal{H}_2$ optimal state feedback controllers can be parameterized as $K_{opt} = \mathcal{F}_\ell(M_{sf}, \Theta)$ with $\Theta \in \mathcal{RH}_2$ and*

$$M_{sf} = \begin{bmatrix} F_2 & I \\ V_c^*(sI - A_{F_2}) & -V_c^* B_2 \end{bmatrix}.$$

Proof. Since

$$T_{vw} = \left(I - \left[\begin{array}{c|c} A_{F_2} & B_2 \\ \hline K - F_2 & 0 \end{array}\right] \right)^{-1} \left[\begin{array}{c|c} A_{F_2} & B_1 \\ \hline K - F_2 & 0 \end{array}\right] = 0,$$

we get

$$(K - F_2)(sI - A_{F_2})^{-1} B_1 = 0 \tag{14.17}$$

which is achieved if $K = F_2$. Clearly, this is the only solution if B_1 is square and nonsingular or if K is restricted to be constant and (A_{F_2}, B_1) is controllable.

To parameterize all elements in the equivalence class ($T_{vw} = 0$) to which $K = F_2$ belongs, let

$$P_c(s) := \left[\begin{array}{c|c} A_{F_2} & B_2 \\ \hline I & 0 \end{array}\right].$$

Then all state feedback controllers stabilizing G can be parameterized as

$$K(s) = F_2 + (I + QP_c)^{-1}Q, \quad Q(s) \in \mathcal{RH}_\infty$$

where Q is free. Substitute K in equation (14.17), and we get

$$Q(s)(sI - A_{F_2})^{-1}B_1 = 0. \tag{14.18}$$

Hence we have

$$Q(s)(sI - A_{F_2})^{-1} = \Theta(s)V_c^*, \quad \Theta(s) \in \mathcal{RH}_2.$$

Thus all $Q(s) \in \mathcal{RH}_\infty$ satisfying (14.18) can be written as

$$Q(s) = \Theta(s)V_c^*(sI - A_{F_2}), \quad \Theta(s) \in \mathcal{RH}_2. \tag{14.19}$$

Therefore,

$$\begin{aligned} K_{opt}(s) &= F_2 + (I + \Theta(s)V_c^*(sI - A_{F_2})P_c)^{-1}\,\Theta(s)V_c^*(sI - A_{F_2}) \\ &= F_2 + (I + \Theta(s)V_c^*B_2)^{-1}\,\Theta(s)V_c^*(sI - A_{F_2}), \quad \Theta(s) \in \mathcal{RH}_2 \end{aligned}$$

parameterizes the equivalence class of $K = F_2$. □

14.8.2 Full Information and Other Special Problems

We shall consider the FI problem first.

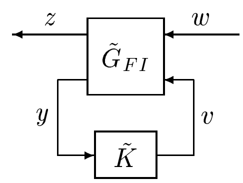
$$G_{FI}(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ \begin{bmatrix} I \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ I \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{array}\right]$$

The assumptions relevant to the FI problem are the same as the state feedback problem. This is similar to the state feedback problem except that the controller now has more information (w). However, as was pointed out in the discussion of the state feedback problem, this extra information is not used by the optimal controller.

Full Information:

1. $\min \|T_{zw}\|_2 = \|G_cB_1\|_2 = (\text{trace}(B_1^*X_2B_1))^{1/2}$
2. $K(s) \cong \begin{bmatrix} F_2 & 0 \end{bmatrix}$
3. $K(s) \cong \begin{bmatrix} F_2 & Q(s) \end{bmatrix}$, *where* $Q \in \mathcal{RH}_2$, $\|Q\|_2^2 < \gamma^2 - \|G_cB_1\|_2^2$

Proof. Items 1 and 2 follow immediately from the proof of the state feedback results because the argument that $T_{vw} = 0$ did not depend on the restriction to state feedback only. Thus we only need to prove item 3. Let K be an admissible controller such that $\|T_{zw}\|_2 < \gamma$. As in the SF proof, define a new control variable $v = u - F_2 x$; then the closed-loop system is as shown below



with

$$\tilde{G}_{FI} = \left[\begin{array}{c|cc} A_{F_2} & B_1 & B_2 \\ \hline C_{1F_2} & 0 & D_{12} \\ \begin{bmatrix} I \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ I \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{array}\right], \quad \tilde{K} = K - [F_2 \ 0].$$

Denote by Q the transfer matrix from w to v; it belongs to $\mathcal{RH}_2$ by internal stability and the fact that D_{12} has full column rank and T_{zw} with $z = C_{1F_2}x + D_{12}Qw$ has finite $\mathcal{H}_2$ norm. Then $u = F_2 x + v = F_2 x + Qw = \begin{bmatrix} F_2 & Q \end{bmatrix} y$ so $K \cong \begin{bmatrix} F_2 & Q \end{bmatrix}$, and $\|T_{zw}\|_2^2 = \|G_c B_1 + UQ\|_2^2 = \|G_c B_1\|_2^2 + \|Q\|_2^2$; hence,

$$\|Q\|_2^2 = \|T_{zw}\|_2^2 - \|G_c B_1\|_2^2 < \gamma^2 - \|G_c B_1\|_2^2.$$

Likewise, one can show that every controller of the form given in item no.3 is admissible and suboptimal. □

The results for DF, OI, FC, and OE follow from the parallel development of Chapter 12.

Disturbance Feedforward:

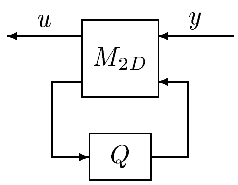
$$G_{DF}(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ C_2 & I & 0 \end{array}\right]$$

This problem inherits the same assumptions (i)-(iii) as in the state feedback problem, in addition to the stability condition of $A - B_1 C_2$.

1. $\min \|T_{zw}\|_2 = \|G_c B_1\|_2$

2. $K(s) = \left[\begin{array}{c|c} A + B_2 F_2 - B_1 C_2 & B_1 \\ \hline F_2 & 0 \end{array}\right]$

3. the set of all transfer matrices from y to u in



$$M_{2D}(s) = \left[\begin{array}{c|cc} A + B_2F_2 - B_1C_2 & B_1 & B_2 \\ \hline F_2 & 0 & I \\ -C_2 & I & 0 \end{array}\right]$$

where $Q \in \mathcal{RH}_2$, $\|Q\|_2^2 < \gamma^2 - \|G_cB_1\|_2^2$

Output Injection:

$$G_{OI}(s) = \left[\begin{array}{c|cc} A & B_1 & I \\ \hline C_1 & 0 & 0 \\ C_2 & D_{21} & 0 \end{array}\right]$$

with the following assumptions:

(i) (C_2, A) *is detectable;*

(ii) D_{21} *has full row rank with* $\begin{bmatrix} D_{21} \\ \tilde{D}_\perp \end{bmatrix}$ *unitary;*

(iii) $\begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix}$ *has full row rank for all ω.*

1. $\min \|T_{zw}\|_2 = \|C_1G_f\|_2 = (\mathrm{trace}(C_1Y_2C_1^*))^{1/2}$

2. $K(s) \cong \begin{bmatrix} L_2 \\ 0 \end{bmatrix}$

Full Control:

$$G_{FC}(s) = \left[\begin{array}{c|cc} A & B_1 & \begin{bmatrix} I & 0 \end{bmatrix} \\ \hline C_1 & 0 & \begin{bmatrix} 0 & I \end{bmatrix} \\ C_2 & D_{21} & \begin{bmatrix} 0 & 0 \end{bmatrix} \end{array}\right]$$

with the same assumptions as an output injection problem.

1. $\min \|T_{zw}\|_2 = \|C_1G_f\|_2 = (\mathrm{trace}(C_1Y_2C_1^*))^{1/2}$

2. $K(s) \cong \begin{bmatrix} L_2 \\ 0 \end{bmatrix}$

3. $K(s) \cong \begin{bmatrix} L_2 \\ Q(s) \end{bmatrix}$, *where* $Q \in \mathcal{RH}_2$, $\|Q\|_2^2 < \gamma^2 - \|C_1 G_f\|_2^2$

Output Estimation:

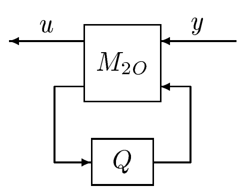
$$G_{OE}(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & I \\ C_2 & D_{21} & 0 \end{array}\right]$$

The assumptions are taken to be those in the output injection problem plus an additional assumption that $A - B_2 C_1$ *is stable.*

1. $\min \|T_{zw}\|_2 = \|C_1 G_f\|_2$

2. $K(s) = \left[\begin{array}{c|c} A + L_2 C_2 - B_2 C_1 & L_2 \\ \hline C_1 & 0 \end{array}\right]$

3. the set of all transfer matrices from y *to* u *in*



$$M_{2O}(s) = \left[\begin{array}{c|cc} A + L_2 C_2 - B_2 C_1 & L_2 & -B_2 \\ \hline C_1 & 0 & I \\ C_2 & I & 0 \end{array}\right]$$

where $Q \in \mathcal{RH}_2$, $\|Q\|_2^2 < \gamma^2 - \|C_1 G_f\|_2^2$

14.9 Separation Theory

Given the results for the special problems, we can now prove Theorem 14.7 using separation arguments. This essentially involves reducing the output feedback problem to a combination of the Full Information and the Output Estimation problems.

14.9.1 $\mathcal{H}_2$ Controller Structure

Recall that the unique $\mathcal{H}_2$ optimal controller is

$$K_2(s) := \left[\begin{array}{c|c} \hat{A}_2 & -L_2 \\ \hline F_2 & 0 \end{array}\right] = \left[\begin{array}{c|c} A + B_2 F_2 + L_2 C_2 & Y_2 C_2^* \\ \hline -B_2^* X_2 & 0 \end{array}\right]$$

and $$\min \|T_{zw}\|_2^2 = \|G_c B_1\|_2^2 + \|F_2 G_f\|_2^2$$

where $X_2 := Ric(H_2)$ and $Y_2 := Ric(J_2)$ and the min is over all stabilizing controllers. Note that F_2 is the optimal state feedback in the Full Information problem and L_2 is the optimal output injection in the Full Control case. The well-known separation property of the $\mathcal{H}_2$ solution is reflected in the fact that K_2 is exactly the optimal output estimate of $F_2 x$ and can be obtained by setting $C_1 = F_2$ in OE.2. Also, the minimum cost is the sum of the FI cost (FI.1) and the OE cost for estimating $F_2 x$ (OE.1).

The controller equations can be written in standard observer form as

$$\begin{aligned} \dot{\hat{x}} &= A\hat{x} + B_2 u + L_2(C_2\hat{x} - y) \\ u &= F_2\hat{x} \end{aligned}$$

where $\hat{x}$ is the optimal estimate of x.

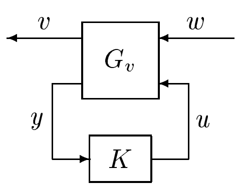
14.9.2 Proof of Theorem 14.7

As before we define a new control variable, $v := u - F_2 x$, and the transfer function to z becomes

$$z = \left[\begin{array}{c|cc} A_{F_2} & B_1 & B_2 \\ \hline C_{1F_2} & 0 & D_{12} \end{array}\right] \begin{bmatrix} w \\ v \end{bmatrix} = G_c B_1 w + U v \tag{14.20}$$

where $G_c(s) := \left[\begin{array}{c|c} A_{F_2} & I \\ \hline C_{1F_2} & 0 \end{array}\right]$ and $U(s) := \left[\begin{array}{c|c} A_{F_2} & B_2 \\ \hline C_{1F_2} & D_{12} \end{array}\right]$. Furthermore, U is inner (i.e., $U^{\sim}U = I$) and $U^{\sim}G_c$ belongs to $\mathcal{RH}_2^{\perp}$ from Lemma 14.3.

Let K be any admissible controller and notice how v is generated:



$$G_v = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline -F_2 & 0 & I \\ C_2 & D_{21} & 0 \end{array}\right]$$

Note that K stabilizes G iff K stabilizes G_v (the two closed-loop systems have identical A-matrices) and that G_v has the form of the Output Estimation problem. From (14.20) and the properties of U we have that

$$\min \|T_{zw}\|_2^2 = \|G_c B_1\|_2^2 + \min \|T_{vw}\|_2^2. \tag{14.21}$$

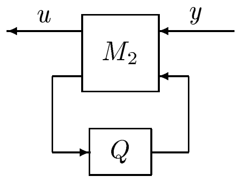
But from item OE.2, $\|T_{vw}\|_2$ is minimized by the controller

$$\left[\begin{array}{c|c} A + B_2F_2 + L_2C_2 & -L_2 \\ \hline F_2 & 0 \end{array}\right],$$

and then from OE.1 $\min \|T_{vw}\|_2 = \|F_2 G_f\|_2$.

14.9.3 Proof of Theorem 14.8

Continuing with the development in the previous proof, we see that the set of all sub-optimal controllers equals the set of all K's such that $\|T_{vw}\|_2^2 < \gamma^2 - \|G_cB_1\|_2^2$. Apply item OE.3 to get that such K's are parameterized by



$$M_2(s) = \left[\begin{array}{c|cc} \hat{A}_2 & -L_2 & B_2 \\ \hline F_2 & 0 & I \\ -C_2 & I & 0 \end{array}\right]$$

with $Q \in \mathcal{RH}_2$, $\|Q\|_2^2 < \gamma^2 - \|G_cB_1\|_2^2 - \|F_2G_f\|_2^2$.

14.10 Stability Margins of $\mathcal{H}_2$ Controllers

We have shown that the system with LQR controller has at least 60^o phase margin and $6dB$ gain margin. However, it is not clear whether these stability margins will be preserved if the states are not available and the output feedback $\mathcal{H}_2$ (or LQG) controller has to be used. The answer is provided here through a counterexample: there are no guaranteed stability margins for a $\mathcal{H}_2$ controller.

Consider a single input and single output two state generalized dynamical system:

$$G(s) = \left[\begin{array}{c|cc} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} & \begin{bmatrix} \sqrt{\sigma} & 0 \\ \sqrt{\sigma} & 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \hline \begin{bmatrix} \sqrt{q} & \sqrt{q} \\ 0 & 0 \end{bmatrix} & 0 & \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 1 \end{bmatrix} & 0 \end{array}\right].$$

It can be shown analytically that

$$X_2 = \begin{bmatrix} 2\alpha & \alpha \\ \alpha & \alpha \end{bmatrix}, \quad Y_2 = \begin{bmatrix} 2\beta & \beta \\ \beta & \beta \end{bmatrix}$$

and

$$F_2 = -\alpha \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad L_2 = -\beta \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

where

$$\alpha = 2 + \sqrt{4+q}\ , \quad \beta = 2 + \sqrt{4+\sigma}.$$

Then the optimal output $\mathcal{H}_2$ controller is given by

$$K_{opt} = \left[\begin{array}{cc|c} 1-\beta & 1 & \beta \\ -(\alpha+\beta) & 1-\alpha & \beta \\ \hline -\alpha & -\alpha & 0 \end{array}\right].$$

Suppose that the resulting closed-loop controller (or plant G_{22}) has a scalar gain k with a nominal value $k = 1$. Then the controller implemented in the system is actually

$$K = kK_{opt},$$

and the closed-loop system A-matrix becomes

$$\tilde{A} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & -k\alpha & -k\alpha \\ \beta & 0 & 1-\beta & 1 \\ \beta & 0 & -\alpha-\beta & 1-\alpha \end{bmatrix}.$$

It can be shown that the characteristic polynomial has the form

$$\det(sI - \tilde{A}) = a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0$$

with

$$a_1 = \alpha + \beta - 4 + 2(k-1)\alpha\beta, \quad a_0 = 1 + (1-k)\alpha\beta.$$

Note that for closed-loop stability it is necessary to have $a_0 > 0$ and $a_1 > 0$. Note also that $a_0 \approx (1-k)\alpha\beta$ and $a_1 \approx 2(k-1)\alpha\beta$ for sufficiently large α and β if $k \neq 1$. It is easy to see that for sufficiently large α and β (or q and σ), the system is unstable for arbitrarily small perturbations in k in either direction. Thus, by choice of q and σ, the gain margins may be made arbitrarily small.

It is interesting to note that the margins deteriorate as control weight $(1/q)$ gets small (large q) and/or system deriving noise gets large (large σ). In modern control folklore, these have often been considered ad hoc means of improving sensitivity.

It is also important to recognize that vanishing margins are not only associated with open-loop unstable systems. It is easy to construct minimum phase, open-loop stable counterexamples for which the margins are arbitrarily small.

The point of these examples is that $\mathcal{H}_2$ (LQG) solutions, unlike LQR solutions, provide no global system-independent guaranteed robustness properties. Like their more classical colleagues, modern LQG designers are obliged to test their margins for each specific design.

It may, however, be possible to improve the robustness of a given design by relaxing the optimality of the filter (or FC controller) with respect to error properties. A successful approach in this direction is the so called LQG loop transfer recovery (LQG/LTR) design technique. The idea is to design a filtering gain (or FC control law) in such way so that the LQG (or $\mathcal{H}_2$) control law will approximate the loop properties of the regular LQR control. This will not be explored further here; interested reader may consult related references.

14.11 Notes and References

The detailed treatment of $\mathcal{H}_2$ related theory, LQ optimal control, Kalman filtering, etc., can be found in Anderson and Moore [1990] or Kwakernaak and Sivan [1972].